

Regulatory & Risk Framework Primer for Quants

The Hidden Geometry of Capital and Model Design

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January 17, 2026

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Core Formula

$$CVA \approx (1 - R) \sum_{i=1}^n EE(t_i) \times PD(t_{i-1}, t_i)$$

The Tenor Gravity

Core Formula

$$CVA \text{ Capital} \propto \text{Duration} \times \text{Expected Exposure}$$

Direct Relationship: Long-dated swaps have much higher CVA capital because the 'time at risk' for credit spread widening is greater. Tenor is the primary driver of the 'Multiplier' effect in XVA desks.

The 3-Second Answer (Module 7)

Question: "Explain CVA to a non-quant trader."

Your opening line:

CVA is the 'Bad Debt Reserve' for your derivative book. It's the market price you'd pay to hedge the risk that your counterparty goes bankrupt while they owe you money.

where:

- R : Recovery rate.
- $EE(t_i)$: Expected Exposure at time t_i .
- $PD(t_{i-1}, t_i)$: Probability of default between t_{i-1} and t_i .

Practitioner Insight

The Computational Burden: Notice that CVA requires calculating **Expected Exposure (EE)***, which is an average over many Monte Carlo paths. **CVA Capital** then requires calculating the **Sensitivity** of this average to credit spreads. This is a "nested" problem: you need a Monte Carlo simulation just to get the price, and another layer to get the capital. This is why XVA desks are the largest consumers of GPU compute in a bank.

7.2 Regulatory CVA (SA-CVA)

The Basel III SA-CVA framework prescribes how to aggregate these sensitivities into a capital charge.

Core Formula

$$K = \sqrt{\sum_i K_i^2 + \sum_i \sum_{j \neq i} \rho_{ij} K_i K_j}$$

where K_i represents the capital for a specific risk factor (e.g., EUR 10Y rate delta), and ρ_{ij} is the regulatory correlation.

7.4 The XVA Desk Conflict: Pricing vs Capital

Internal quants calculate "Economic CVA" (best estimate). Regulators mandate "Regulatory CVA" (conservative formulas).

1. **Case Study: The Corporate Swap** If a corporate client doesn't post collateral, the EE remains high for the life of the trade.

The "Documentation Tax"

A junior quant spends 20% of their time coding and 80% of their time writing the MDD. **Pro Tip:** Use automated documentation tools (LaTeX-based) that pull Greeks directly from your Python/C++ test scripts to ensure the math in the document matches the code in production.

Appendix C: Advanced CVA Derivation (Jump-Diffusion)

In Module 7, we assumed credit spreads move smoothly. In reality, they jump.

Core Formula

$$d\lambda_t = \kappa(\theta - \lambda_t)dt + \sigma\sqrt{\lambda_t}dW_t + J_t dq_t$$

where J_t is a jump size distribution. This "Jump-to-Default" risk is why CVA capital is often much higher than the expected loss—the "Tail" of the spread distribution is much fatter than a Normal distribution assumes.

Appendix D: The Quant's Career Path in Regulation

- **Front Office Quant:** Focus on Capital Optimization (minimizing RWA).
- **Risk Quant:** Focus on Methodology (designing the ES / VaR engines).
- **Model Validator (IVU):** Focus on Skepticism (finding the holes in the FO models).

Which pays better? Currently, FO Quants with Regulatory knowledge are the highest in demand, as banks move from "Alpha" generation to "Capital" efficiency.

Stability is key for capital planning.

9. **What is a "Model Reserve"? Ans:** A cash buffer set aside to cover the estimated error of a model. It's often used when a model is known to be slightly inaccurate but building a better one is too expensive or slow.
10. **Define "Governance Committees". Ans:** A group of senior stakeholders (Risk, Finance, Tech) who have the final authority to approve or reject a model's usage for the bank's book.

E.4 Strategic & Desk Impact: Questions & Answers

1. **How does high RWA affect a desk's ROE? Ans:** RWA is the denominator. $ROE = PnL / (8\% \times RWA)$. If RWA doubles, you need to double your PnL just to maintain the same ROE.
2. **Explain "Proxy Hedging" and its capital benefits. Ans:** By hedging an illiquid (NMRF) risk with a liquid (Modellable) proxy, you can move risk from a "Stress Add-on" bucket into the "Internal Model" bucket, significantly lowering capital.
3. **What is "Regulatory Arbitrage"? Ans:** Structuring trades to exploit differences between models. Example: Moving a trade from the Trading Book to the Banking Book to use accrual accounting instead of Mark-to-Market.
4. **How does NSFR affect a Repo desk? Ans:** The Net Stable Funding Ratio forces desks to have long-term funding for long-term assets. This makes "borrowing short to lend long" (the carry trade) much more expensive.
5. **Discuss LCR and HQLA. Ans:** LCR requires banks to have enough "High Quality Liquid Assets" (cash/gov bonds) to survive a 30-day "run on the bank". This makes holding illiquid exotics very expensive for the balance sheet.
6. **Why have many banks exited Exotic Correlation? Ans:** Because of the "Correlation Floor" in Basel IV. Even if you hedge your risk, the regulator forces you to hold capital as if you were unhedged, making the business unprofitable.
7. **How does CVA capital affect swap pricing? Ans:** Banks now charge an "XVA spread" to corporate clients. If the client is uncollateralized, this charge can be larger than the interest rate spread itself.
8. **Explain the "Capital Stack". Ans:** CET1 (Common Equity) is the base; AT1 (Contingent Convertibles) is the middle; Tier 2 (Sub-debt) is the top. CET1 is the most expensive and highest quality.
9. **What is the "Leverage Ratio"? Ans:** A non-risk-based backstop. It's simply $Capital / Total Assets \geq 3\%$. It stops banks from using "Quant Magic" to reduce RWA to near zero.
10. **How has regulation changed the career path of a Quant? Ans:** Quants are no longer just "Math Geeks"; they are now "Strategic Risk Managers". Success requires understanding the SDE as well as the 1,000-page Basel IV rulebook.
11. **Python: How do you implement a 97.5% ES engine using Importance Sampling? Ans:** Since ES integrates the tail, standard MC is slow. Importance Sampling shifts the sampling distribution towards the tail (the "Failure Region") to reduce variance and speed up convergence.
12. **C++: Why is memory alignment critical for CVA engines? Ans:** CVA requires millions of paths. Using SIMD (Single Instruction, Multiple Data) instructions requires data to be memory-aligned. This allows the CPU to process 4-8 prices in a single clock cycle.
13. **Data: How do you handle 'stale prices' in an FRTB backtest? Ans:** Stale prices are a red flag. You must apply a "Proxy" liquid instrument or a "Linear Interpolation" with a documented liquidity haircut to pass the SR 11-7 data quality check.
14. **Hedging: Explain the 'Delta-Neutral, Gamma-Short' trap with barriers. Ans:** Being delta-neutral doesn't save you from a barrier cliff. As you approach the barrier, gamma explodes, and a small move creates a massive delta gap that the desk can't hedge fast enough.